

NGOC MINH TRAN (PhD)

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Nationality: Dual Irish and Vietnamese

PROFESSIONAL SUMMARY

AI Leader with 12+ years of international experience driving enterprise-scale AI Factory initiatives. Currently leading Mastercard's flagship "big bet" project: building and scaling a transaction-based Transformer Foundation Model across hundreds of billions of financial records using H100 GPU clusters. Proven track record in translating deep-tech infrastructure into high-impact banking use cases (SME growth, risk, churn). Expert in AI Security, Federated Learning, and AutoAI with 21 US/EU patents and 16 scientific publications.

CORE COMPETENCIES

- **Generative AI & Infrastructure:** Transformer Architectures, Foundational Models, Large-Scale Distributed Training (H100 Clusters), MLOps at Scale.
- **Enterprise Downstream Use Cases:** Hidden SME Discovery, SME Credit Risk Score, Customer Churn Optimization, Attrition & Reactivation, Expenditure Forecasting, Customer Segmentation.
- **AI Security:** Privacy-Preserving AI, Federated Learning, Adversarial AI.
- **Big Data Engineering:** Enterprise Cloud-Native Infrastructure (Databricks, AWS), Spark, Hadoop.

PROFESSIONAL EXPERIENCE

MASTERCARD | Dublin, Ireland

Director of Data Science / Principal Data Scientist (AI/ML) | 2021 – Present

- **Team & Project Leadership:** Lead a globally distributed team of data scientists across the US and EU; collaborate with leadership, product managers, and multi-functional stakeholders to align AI roadmaps with business KPIs.
- **Foundation Model Architecture:** Architect and build a proprietary transaction-based Foundation Model using Transformer architectures, scaling deep learning training across hundreds of billions of global transactions utilizing multi-node H100 GPU clusters.
- **GenAI & RAG Implementation:** Design and implement robust RAG systems and fine-tune foundational LLMs on internal enterprise data to accelerate cross-functional project onboarding.
- **Banking & Commercial Use Cases:** Develop ML applications directly driving core business metrics, including SME Credit Risk Scores (CRS), Hidden SME Identification, Fraud Detection, Customer Attrition/Reactivation, and Liquidity Forecasting.
- **Financial Optimization:** Optimize multi-million dollar compute budgets and MLOps pipelines on Databricks for ultra-large-scale foundational model deployment.

IBM RESEARCH LAB | Dublin, Ireland

Staff Research Engineer (AI & Cloud) | 2017 – 2021

- **[Adversarial Robustness Toolbox \(ART\)](#):** Served as Architect, Maintainer, and Core Reviewer for the flagship open-source AI security framework funded by the US Defense Advanced Research Projects Agency (DARPA, Contract HR001120C0013); delivered **1,300+ commits** to a platform adopted globally with **6,200+ GitHub stars** and **1,000+ research citations**.
- **[Federated Learning \(IBMFL\)](#):** Co-architected and developed IBM's enterprise and open-source Federated Learning Python frameworks, establishing secure multi-party AI training standards tailored for highly regulated cloud and edge banking environments; project garnered **530+ GitHub stars** and **250+ research citations**.
- **IP Generation & Corporate Value:** Invented and filed **21 US & EU patents** as lead/co-inventor, successfully converting deep-tech research into commercial corporate IP portfolios.

- **Security & AutoAI R&D:** Led advanced R&D initiatives in Enterprise AI Security, protecting production-grade LLM and ML models against systemic vulnerabilities and adversarial attacks.
- **Recognition:** Honored with **4 Outstanding Technical Achievement Awards and Division Awards** for exceptional contributions to scaling Federated AI and Security AI infrastructures.

SILICON VALLEY STARTUP | California, USA (Remote/Onsite)

Senior AI & Big Data Engineer | 2014 – 2016

- **Architected end-to-end Big Data pipelines** using Apache Spark, Hadoop, and HBase to process large-scale data streams.
- **Developed a cloud-based recommendation engine** for TV and movie programs serving ~40 million users.
- **Developed machine learning models** for predictive, commercial applications within manufacturing sectors.

EDUCATION

- **Postdoc in Computer Science** | Institute for Research in Computer Science and Automation (INRIA), France | 2012 – 2013.
- **PhD. in Computer Science** | Leiden University, The Netherlands | 2008 – 2012.

HONORS & ACHIEVEMENTS

Corporate Awards

- [IBM Research Division Award in Security AI 2019.](#)
- [IBM Research Division Award in Auto AI 2019.](#)
- [IBM Outstanding Technical Achievement Award in Security AI 2019.](#)
- [IBM Outstanding Technical Achievement Award in Federated AI for Hybrid Cloud and Edge 2021.](#)
- [IBM First Plateau Invention Achievement Award 2019.](#)
- [IBM Second Plateau Invention Achievement Award 2019.](#)
- [IBM Third Plateau Invention Achievement Award 2020.](#)

Publications & Patents

- **Patents:** Lead inventor (12) and co-inventor (9) of **21 US and EU patents** spanning Security AI, Federated AI, Auto AI, Generative AI, and FinTech AI.
- **Publications:** First author (10) and co-author (6) of **16 scientific papers** in top-tier AI and Cloud Computing venues (Full list via [Google Scholar](#)).

Open Source Recognitions

- [Open Source Strategic Committer in 2019.](#)
- [Pull-Shark achievements 2 times in Github.](#)
- [Arctic Code Vault contributor in Github.](#)

Professional Certifications

- [Developer Certification for Apache Spark](#), certified by Databricks and O'Reilly Media, Jan 2016.
- [Oracle Certified Professional, Java SE 6 Programmer](#), Oct 2015.
- [Revolution R Enterprise Certified Specialist](#), Aug 2015.
- [Cloudera Certified Specialist in Apache HBase](#), Dec 2014.
- [Cloudera Certified Developer for Apache Hadoop](#), Oct 2014.
- [Cloudera Certified Administrator for Apache Hadoop](#), Oct 2014.